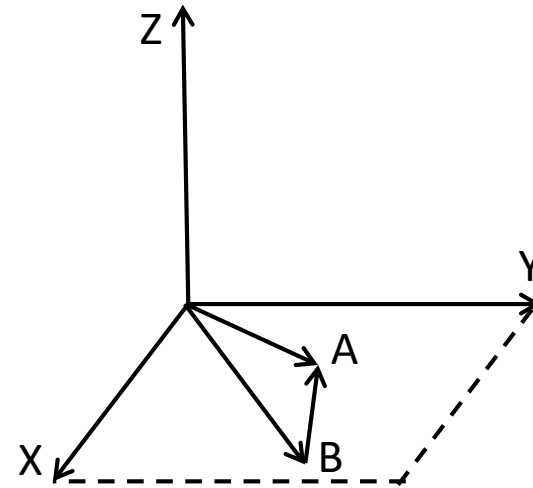


1.

The point A has spherical coordinates $\theta = \pi/2$, $\phi = \pi/3$, $r = 6$; the Cartesian coordinates of the point B are $x = 5$, $y = 5\sqrt{3}$, $z = 0$. Find the distance between the points A and B.

$$A = (3, 3\sqrt{3}, 0)$$

$$B = (5, 5\sqrt{3}, 0)$$

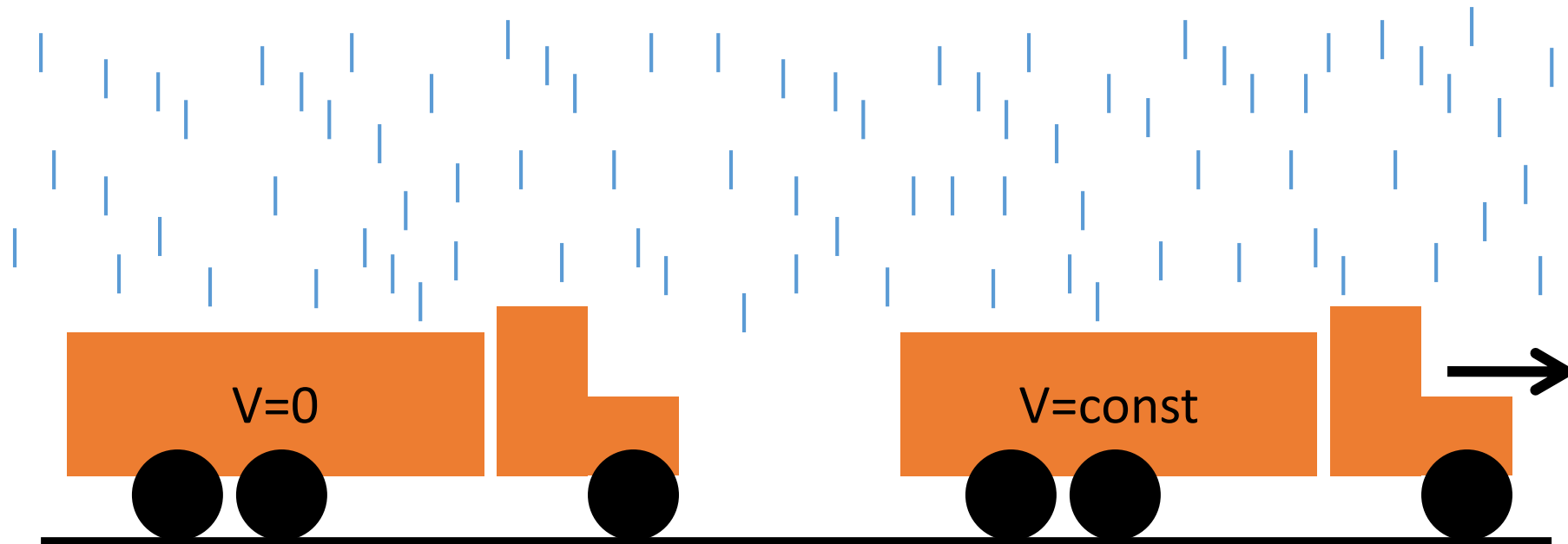


$$|\vec{B} - \vec{A}| = \sqrt{(b_x - a_x)^2 + (b_y - a_y)^2 + (b_z - a_z)^2} = \sqrt{(5 - 3)^2 + (5\sqrt{3} - 3\sqrt{3})^2 + (0 - 0)^2} = 4$$

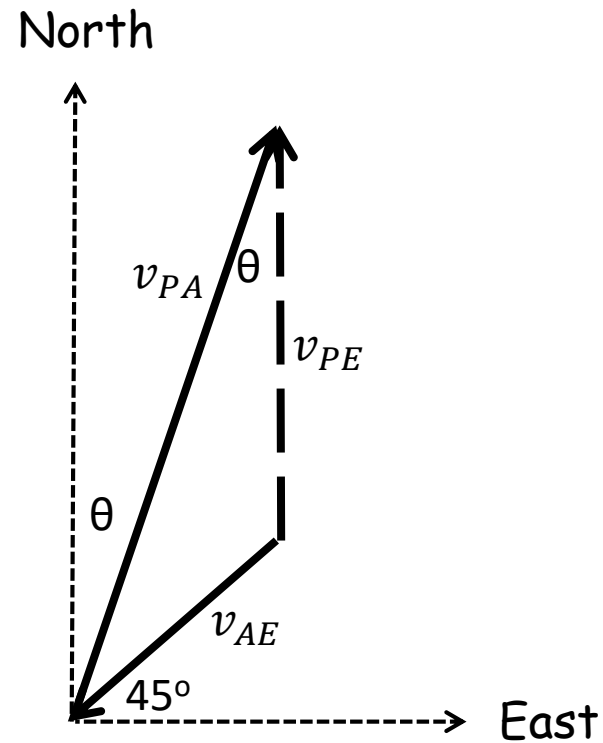
Solution

2.

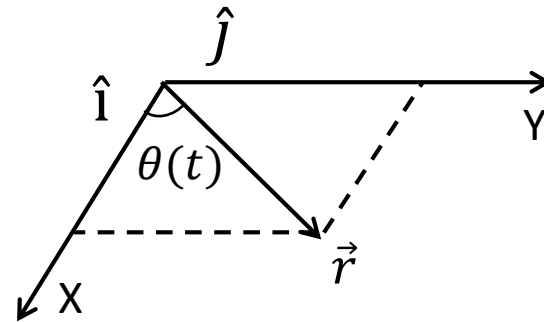
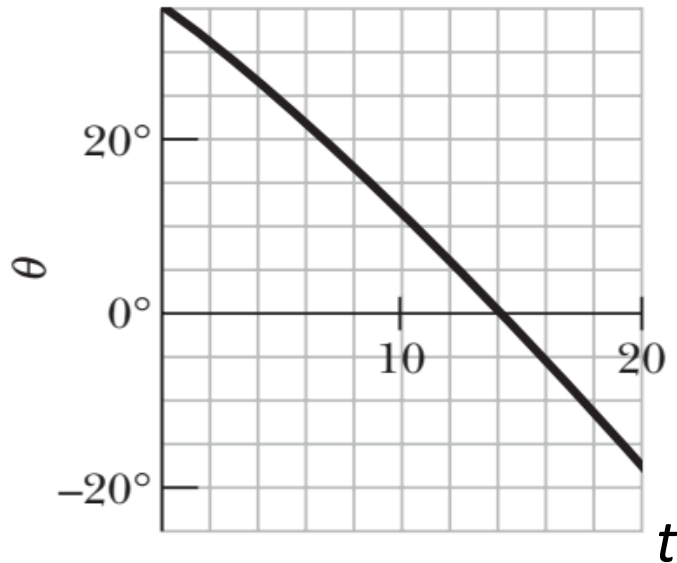
Two identical trucks are caught in the rain. First truck is at rest, second one is moving with constant velocity V . Which truck gets more rain water per minute through the open dump truck body?



3. A pilot with an airspeed (a speed with respect to air) of $v_{PA} = 120$ km/h wishes to fly due to north. A wind $v_{AE} = 40$ km/h is blowing from the northeast. In what direction θ should she head, and what will be her ground speed v_{PE} (a speed with respect to ground)?

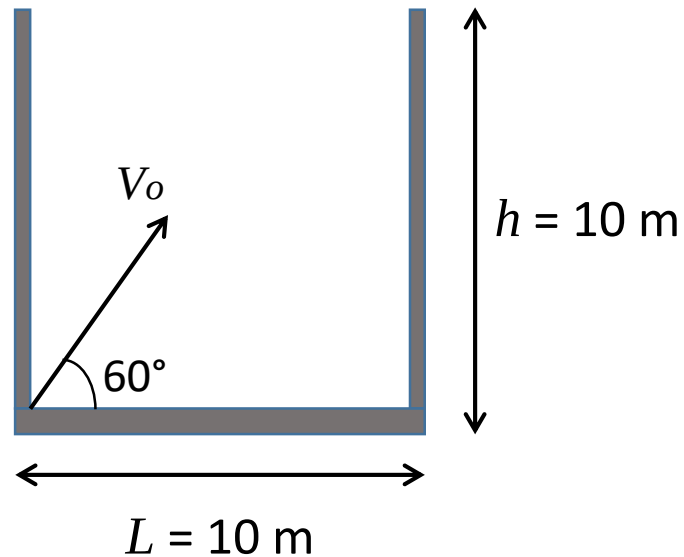


4. The position vector $\vec{r} = 5t \cdot \hat{i} + (et + ft^2)\hat{j}$ locates a particle as a function of time t . The vector $\vec{r}$ is in meters, t is in seconds. The figure gives the angle θ of the position vector's direction as a function of time (θ is measured from the positive x direction). What are the values of the constants e and f ?

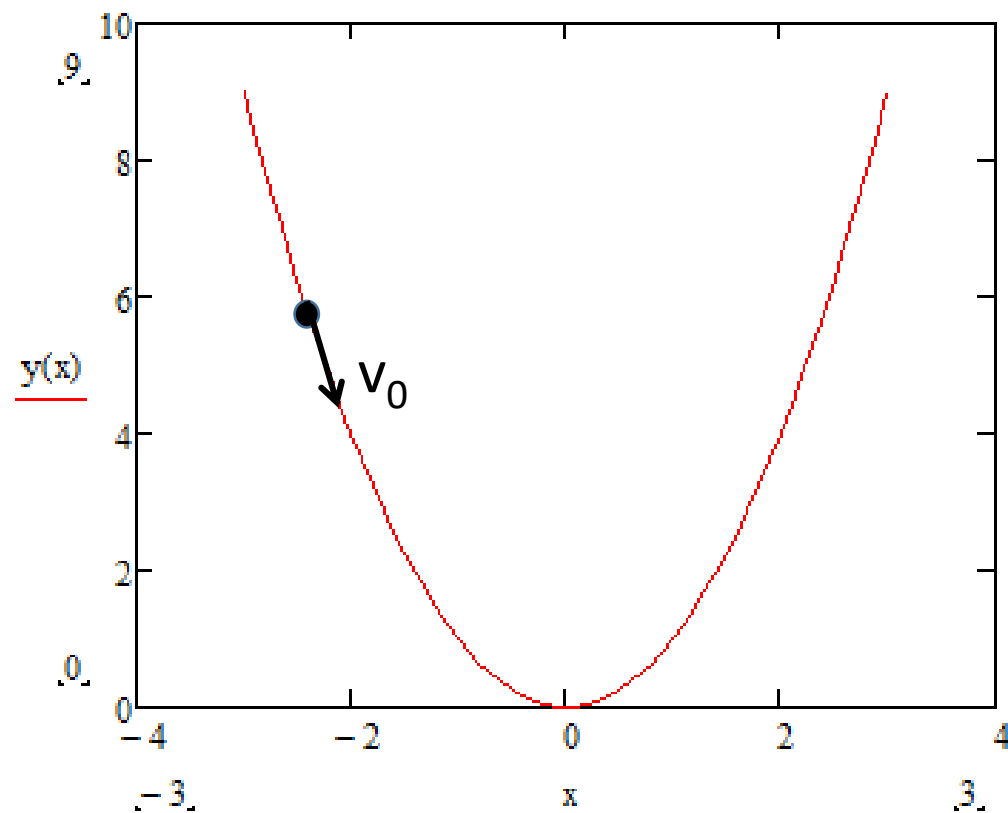


5.

The ball is thrown from the bottom of a rectangular pit with an initial velocity $V_0 = 20$ m/s which direction makes an angle of $\alpha = 60^\circ$ to the horizon line. The pit depth is $h = 10$ m, the distance from the throwing point to the pit wall is $L = 10$ m. Will the ball leave the pit?



6. The material point moves in plane with a constant magnitude of the velocity v_0 . Its trajectory is given by equation $y(x) = c x^2$. Find at the point $x = 0$ the acceleration of the material point and the trajectory radius.



7. A traveler has to get at the final C position from the initial A position via the intermediate B point. His walking speed is 7 km/h and swimming speed is 2 km/h. Find the distance AB, which allows fastest travel from A to C. Take into account $H = 200$ m, $L = 45$ m.

